



Analysis of Instruction-Following Failures in Text-Guided Image Editing

Course: Machine Learning Sessional - CSE 472

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Project Overview: Defining the Problem Space

We are studying **instruction-following failures** in text-guided image editing using models like InstructPix2Pix or Stable Diffusion.

01

Original Image

The unmodified input image serving as the baseline for editing operations.

02

Edit Prompt

Natural language instruction specifying the desired transformation.

03

Edited Image

Model-generated output reflecting the applied edit instruction.

Core Objective: Introduce human-annotated instruction adherence labels and a structured error taxonomy to collect explicit judgments on edit accuracy and analyze correlations between prompt properties and failure modes.



The Challenge in Current Text-Guided Editing

The Issue

Generative models frequently fail to follow specific user instructions accurately. For example, "make the cat red" might change the background instead of the intended subject.

Current Progress

Existing metrics like CLIP score often fail to capture subtle semantic nuances or specific instruction adherence.

Why This Project?

- Lack of granular data explaining why edits fail
- Need to move beyond binary "good/bad" ratings
- Detailed error taxonomy is essential
- Critical for building more robust editing models

Methodology: The Annotation Workflow

Our end-to-end pipeline processes image editing triplets through a structured human annotation protocol, categorizing editing failures into specific types.

Practical Demonstration: A Failure Case

Original Image



Edit Prompt + Edited Image

"Make the persons wear Hawaiian shirts"



Human Annotation

No (Failure)

Error Type: Over-editing - Background changed to beach when only clothing should have been modified

Deep Dive: Error Taxonomy & Statistical Analysis

We apply detailed error type classification when adherence is rated as Partial or No.

Object Errors

- Wrong Object
- Missing Object
- Extra Object

Attribute Errors

- Wrong Color
- Wrong Size
- Incorrect Properties

Spatial & Style

- Location Error
- Style Mismatch

Editing Scope

- Over-editing
- Under-editing
- Collateral Changes

Analysis Plan: Correlating prompt properties (ambiguity, complexity, specificity) and image properties (clutter, occlusion) with specific failure modes.

Dataset Viewer (First 5GB)

Auto-converted to Parquet

API

Embed

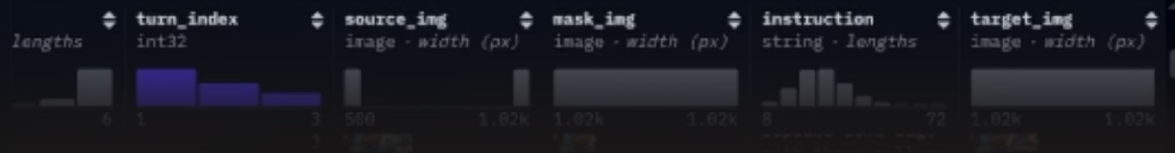
Duplicate

Data Studio

Split (2)

train · 1.73k rows

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⋮

Homepage:

osu-nlp-group.github.io

Repository:

github.com

Point of Contact:

Kai Zhang

Datasets & Resource Availability

Base Dataset

We utilize **MagicBrush** (osunlp/MagicBrush on Hugging Face) as the source for image-editing triplets.

Our Contribution

We extend this dataset by adding a novel layer of human annotations regarding instruction adherence and error type labels not present in the original dataset.

1

Input

Original Image + Edit Prompt

2

Output

Edited Image

3

Label

Adherence Score + Error Categories

Future Work: Automatic Error Prediction

We explore the feasibility of deep learning components for automatic quality assessment.



Output Evaluation Model

Train a classifier to automatically predict "Success/Failure" and error types from triplets.



Difficulty Prediction

Analyze original image and prompt to predict risk of failure before editing occurs.

Potential Use Cases: Automatic quality estimation and prompt warnings for improved user experience.

That's all from us. Thanks a lot.